

HOST HABITAT SPECIALIZATION DETERMINES WHERE MULTIHOST PARASITE SHARING OCCURS ALONG FOREST-AG BOUNDARIES

Authors

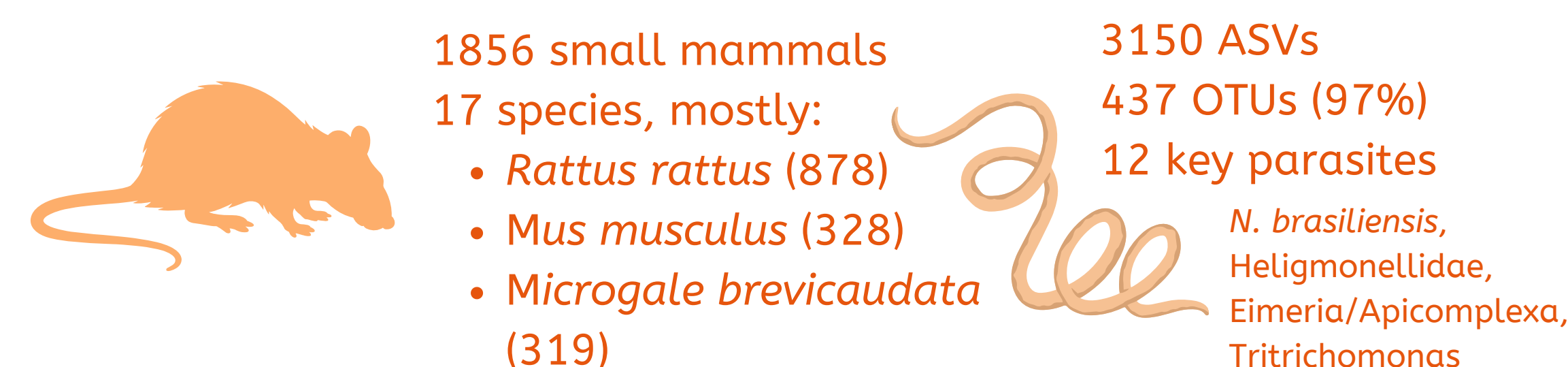
Kayla M. Kauffman^a, Maheriniaina Toky Randriamoria^b, Rianja Randriamifidisoa^b, Jeanne A. Rajaonarivelo^c, Pablo Tortosa^c, Voahangy Soarimalala^{b,d}, Steven M. Goodman^{b,e}, Peter J. Mucha^f, Charles L. Nunn^g, Hillary Young^a, Georgia Titcomb^h

Affiliations

^aUniversity of California Santa Barbara; ^bAssociation Vahatra; ^cUMR PIMIT, Université de La Réunion; ^dUniversité de Fianarantsoa; ^eField Museum of Natural History, Chicago; ^fDartmouth College; ^gDuke University; ^hColorado State University

Introduction

Identifying environmental conditions that facilitate multihost parasite sharing is necessary to help identify critical precursors to disease emergence [1–3]. Parasites found primarily in one host species (primary host) and less commonly in other “secondary host” species are particularly useful for investigating patterns of parasite sharing. Ecosystem boundaries are hypothesized to promote multispecies parasite sharing because species occupying adjacent habitats come into contact at habitat edges [4–6].



Methods

- 1 Trap small mammals (Muridae, Soricidae, Tenrecidae, Nesomyidae) and bats in protected forests, agricultural land use types, & homes.
- 2 Fecal sample metabarcoding to test for a suite of parasites (18S rRNA eukaryotic & ITS-2 nematodes) [7,8]
- 3 Identify key parasites found in >10% of a highly abundant primary host species (*Rattus rattus*, *Mus musculus*, *Mircogale brevicaudata*) and less commonly in other secondary host species (families include: Soricidae, Tenrecidae, Nesomyidae)
- 4 Build species distribution models for primary host occurrence and primary host infection using tidysdm [9] with environmental predictors including cover, NDVI, elevation, aspect, core area, distance to village, etc
- 5 Bayesian hierarchical models of infection in secondary host species based on P(primary occurs), P(primary infected), forest edge distance + random effects for parasite, secondary host taxonomy, and spatial autocorrelation using brms [10]

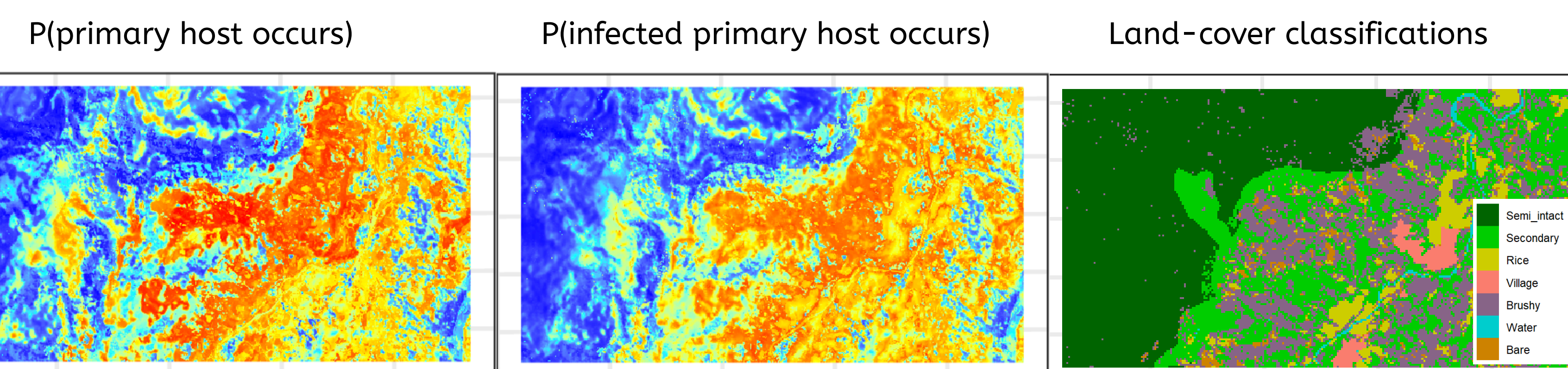


Figure 1. Species distribution model of (A) *Rattus rattus* and (B) *Rattus rattus* infected with *Nippostrongylus brasiliensis* based on environmental predictors, including land cover classifications (C) in 1 of 3 villages.

Hypotheses

Infected secondary hosts are more likely to occur where:

- The probability of primary hosts occurring and/or being infected if highest due to density dependence in transmission
- At forest-matrix edges, because species interact near habitat edges

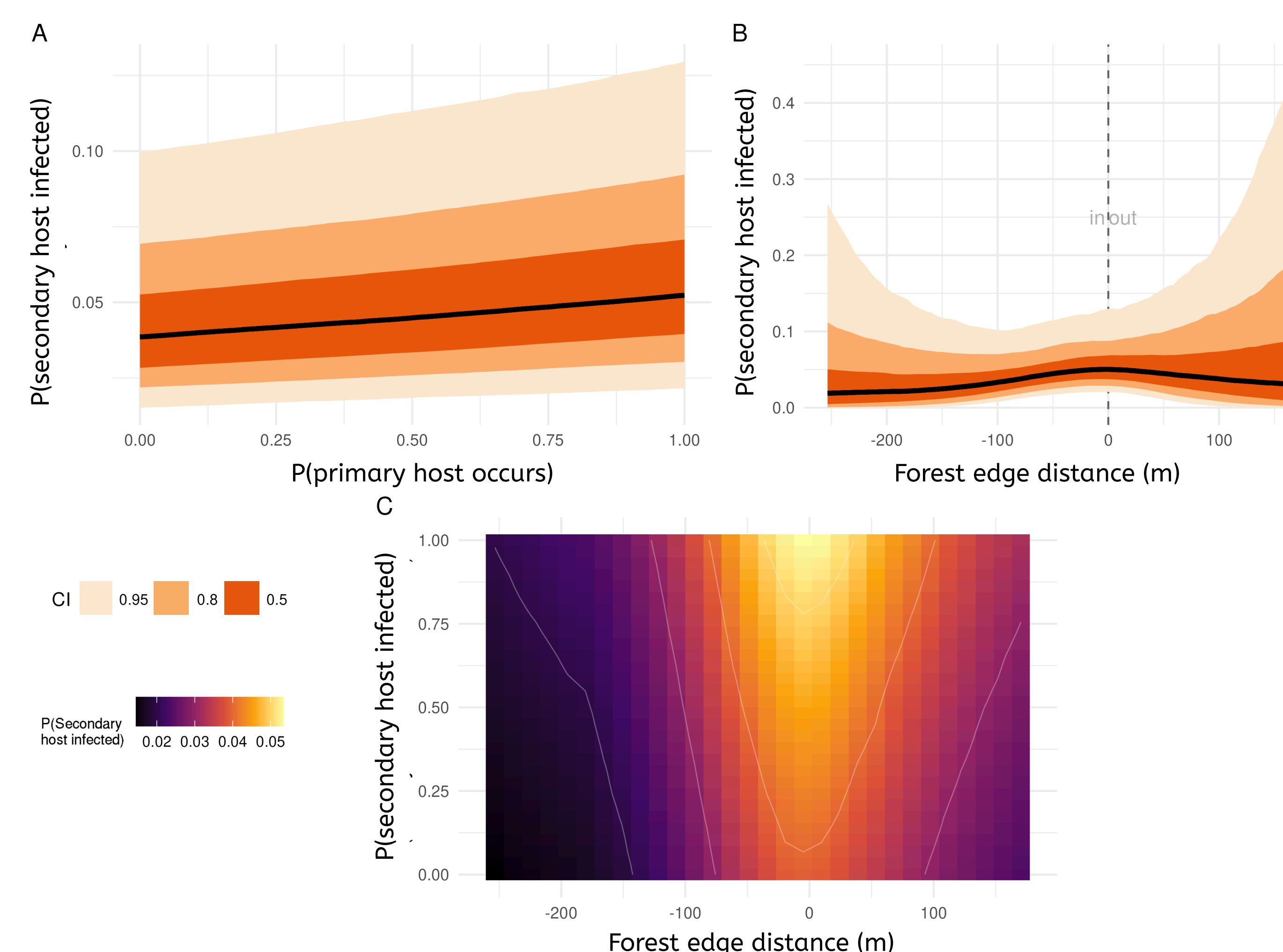


Figure 2. Global effects of the hierarchical model of the probability of infection in secondary host species.

Results

Overall, species distribution models of primary host species and infected primary host species (Figure 1) were similar:

- Pearson’s cor RR:0.87, MM:0.97, Mb:0.93

Global effects across all secondary host species – parasite combinations (Figure 2) show that the probability of infection in secondary hosts increases with:

- P(primary host occurs)
- Near forest edges

The conditional effects (secondary species – parasite) show heterogeneity in this response, with 3 typical responses highlighted in Figure 3.

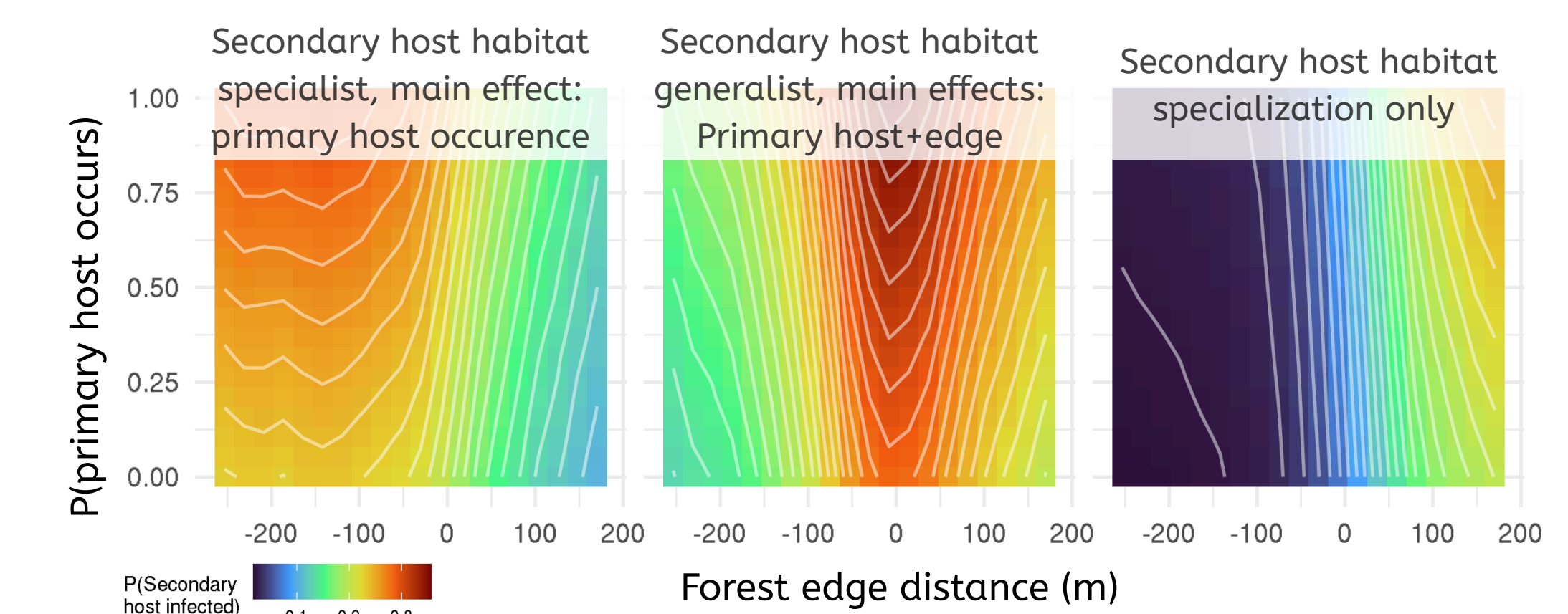


Figure 3. Typical conditional effects of species-parasite combinations.

Conclusions

Combining infection and environmental data across a suite of hosts and parasites showed that multihost parasite sharing is likely to occur at:

- Forest-matrix boundaries
- Where primary hosts are likely to occur.

Effects were highly structured by secondary host-habitat specialization.

- Secondary host species that occurred even rarely in multiple habitat types or near forest-matrix edges were more likely to be infected at those edges than species found only in habitat interiors.

Key Sources & Acknowledgements

[1] Brennan, et al. (2024). Estimating pathogen-spillover risk using host-ectoparasite interactions. *Ecol. Evol.* [2] Kreuder Johnson, et al. (2015). Spillover and pandemic properties of zoonotic viruses with high host plasticity. *Sci. Rep.* [3] Oliva, et al. (2017). Host and viral traits predict zoonotic spillover from mammals. *Nature*. [4] Barremans, et al. (2019). Cross-species pathogen spillover across ecosystem boundaries: mechanisms and theory. *Philos. Trans. R. Soc. B Biol. Sci.* [5] Patz, et al. (2004). Unhealthy Landscapes: Policy Recommendations on Land Use Change and Infectious Disease Emergence. *Environ. Health Perspect.* [6] Ries, et al. (2004). Ecological Responses to Habitat Edges: Mechanisms, Models, and Variability Explained. *Annu. Rev. Ecol. Syst.* [7] Gogarten, et al. (2020). Metabarcoding of eukaryotic parasite communities describes diverse parasite assemblages spanning the primate phylogeny. *Mol. Ecol. Resour.* [8] Gasser, et al. (1993). Rapid sequencing of rDNA from single worms and eggs of parasitic helminths. *Nucleic Acids Res.* [9] Leonard, et al. (2025). Species Distribution Models with Tidymodels. [10] Bürkner, P.-C. (2017). brms: An R Package for Bayesian Multilevel Models Using Stan. *J. Stat. Softw.*

This work was funded by an award from the **NIH-NSF-NIFA Ecology and Evolution of Infectious Disease** program (R01-TW011493), a **Provost’s Collaboratory** from Duke University, two BASS connection grants from Duke University, and the **FACE Foundation**. Thank you to the Duke Lemur Center SAVA Conservation, Association Vahatra, Mention Zoologie et Biodiversité Animale of the Université d’Antananarivo, Madagascar National Parks, and the Ministère de l’Environnement et du Développement Durable, trapping assistants (Daniel Falinirintsoa, Johanna Rafanomezanjanahary, and Lomeris Todilaly), and laboratory interns (Luce Muzi and Alice Tocy).